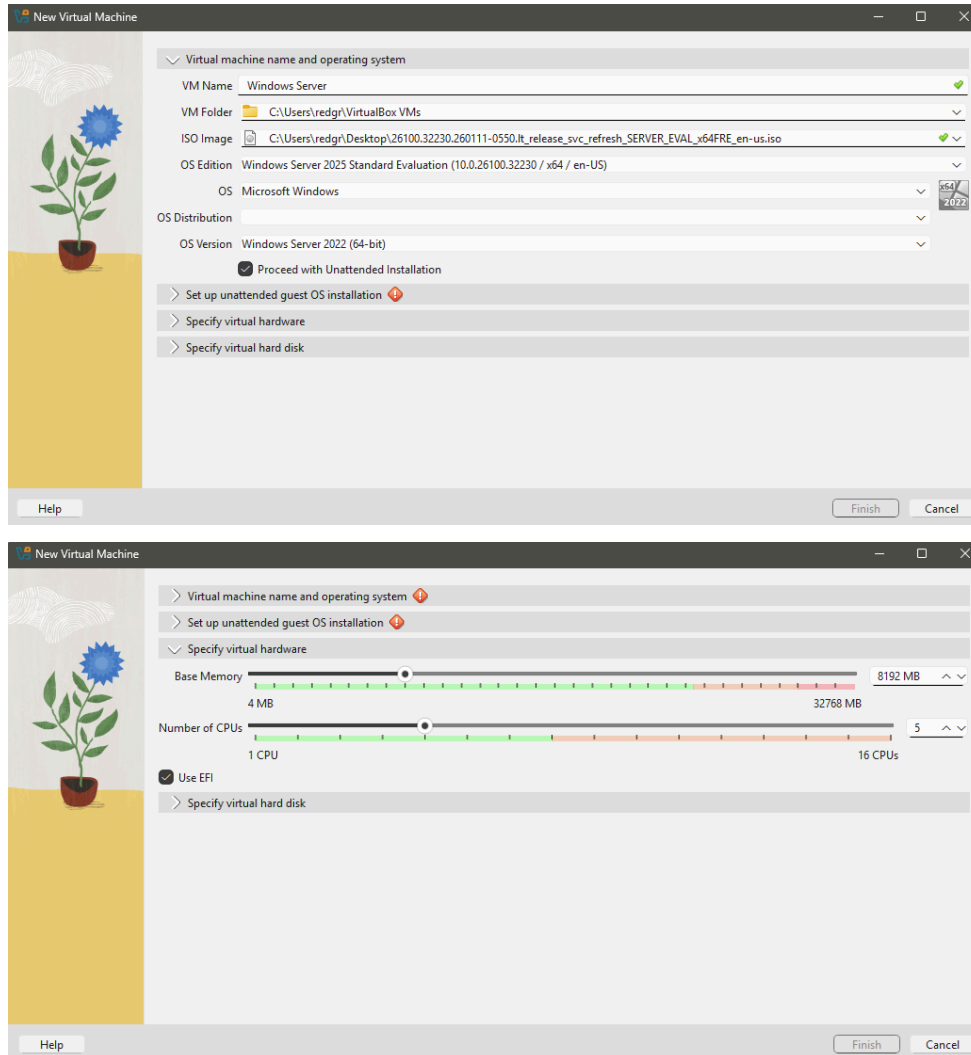


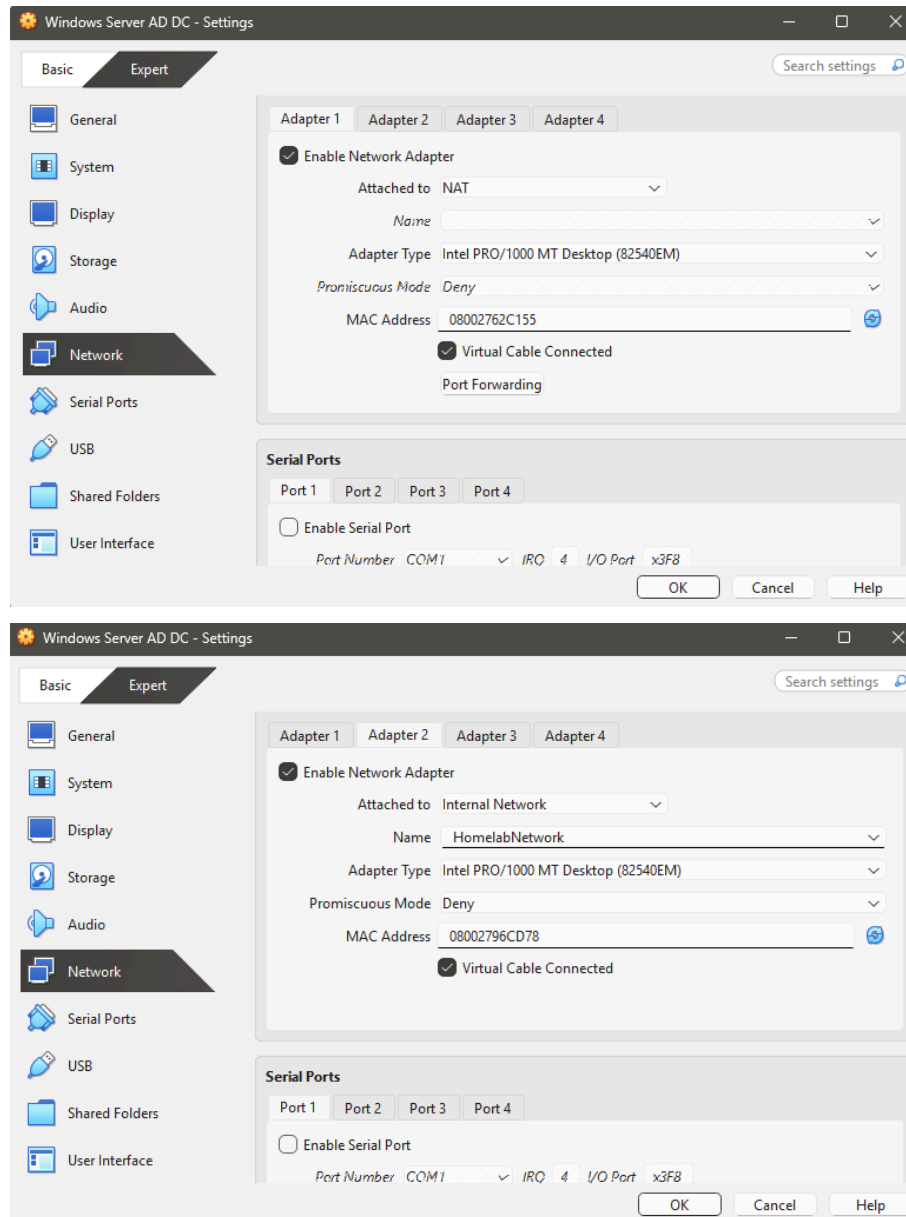
VirtualBox Active Directory Homelab Documentation

Objective 1: Set up a collection of VMs by loading in their ISO images. One VM running Windows Server 2022 to act as the domain controller, followed by 4 VMs running Windows 11 Enterprise to act as client machines. Memory and CPU allocation was increased to approximately 8000MB of memory and 5 CPUs to increase performance.



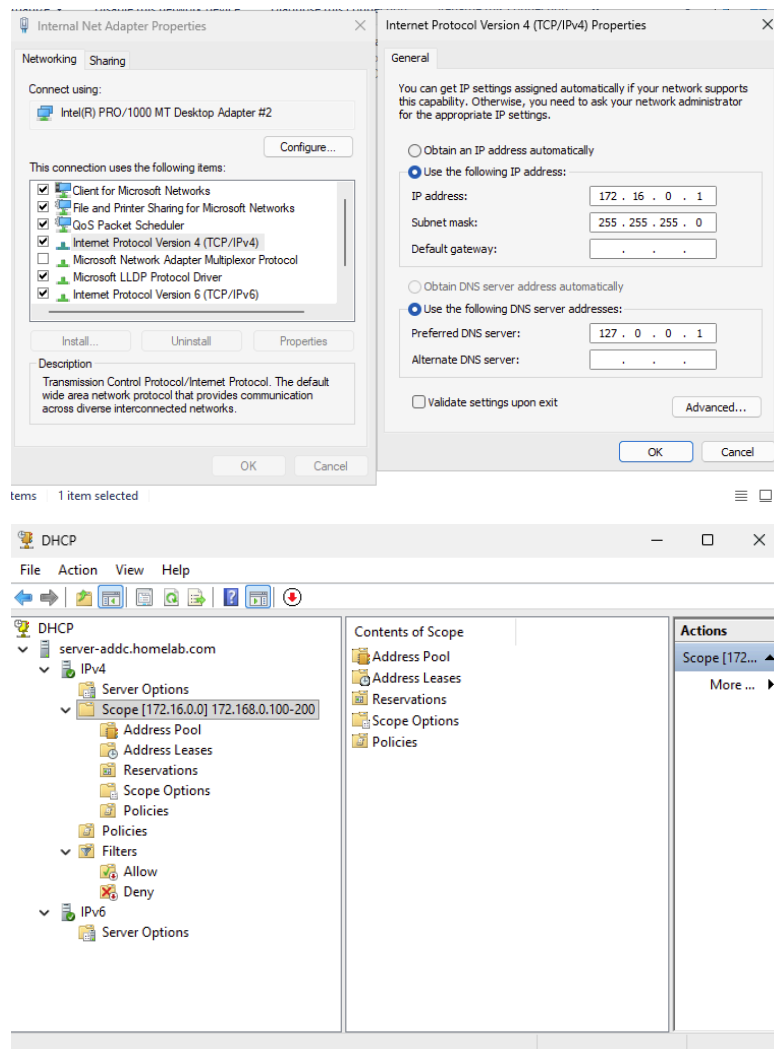
Challenges & Resolution: I was initially, and accidentally, setting up Windows Server to be command line only. Having never used a server OS before, I became quite confused and thought server OSes must be very barebones, which didn't seem right. After some reinstallations, I learned there is a GUI and a command line installation of Windows Server. I opted for GUI and solved the problem.

Objective 2: Customize network adapter settings on the VMs so that they can 1) communicate with the outside Internet, and 2) communicate on an internal network with each other. Each VM was given a NAT network adapter for Internet access, followed by a second Internal Network adapter under the “HomelabNetwork” network for internal communication.



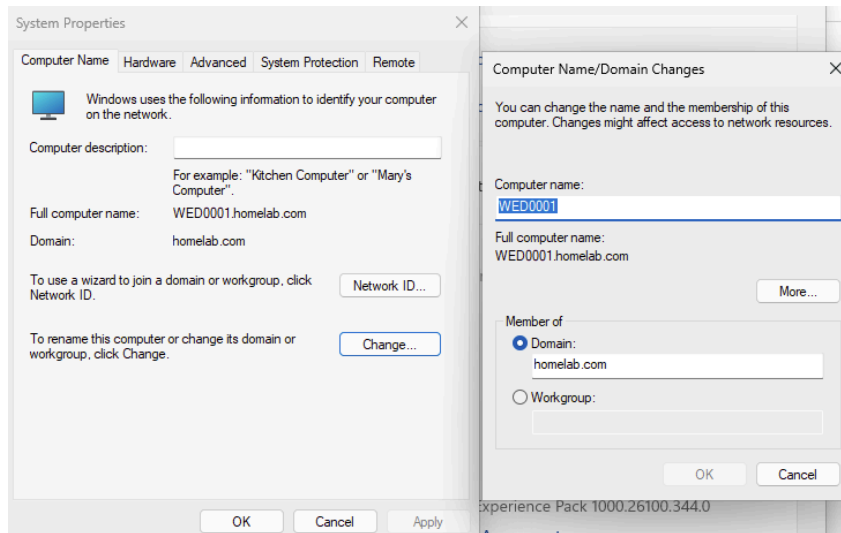
Challenges & Resolution: On my first attempt at setting up this array of VMs, I could get the VMs to talk to each other, but I could not get them to talk to the Internet. This was due to the knowledge source I was using only setting up an Internal Network adapter and nothing else. I researched other sources and learned that 2 network adapters are needed, which I promptly set up to regain Internet access.

Objective 3: Position the Windows Server as the domain controller. This involved setting the host name, changing the internal network adapter settings to give the server a static IP, and configuring the local server with roles and features. Active Directory Domain Services, DHCP, and DNS were installed as server features. The domain was established as “homelab.com”. The DHCP server was configured with a scope of addresses to hand out to devices on this network.

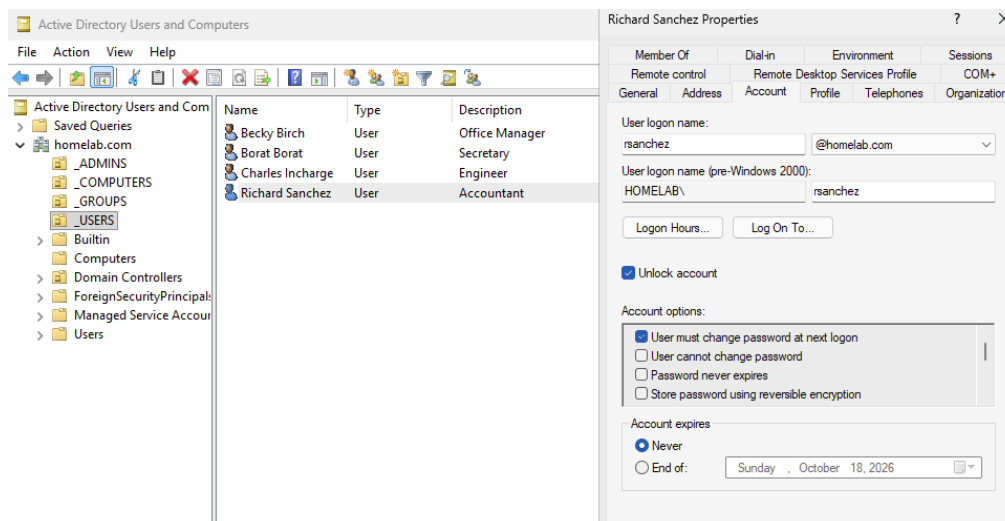


Challenges & Resolution: Another oversight of the previous knowledge source I was using, I did not know I needed a DHCP or DNS server. They were set up at a later point after I learned I would need them for a functional network instead of a closed off Active Directory network with no real use. This drives home the value in researching multiple sources.

Objective 4: Joining the Windows 11 Enterprise VMs to the domain. I followed the path Settings>System Properties>About>Domain or workgroup to access the dialog to add each VM to the homelab.com domain. My DNS, DHCP, and AD servers were now all working together with my VMs.

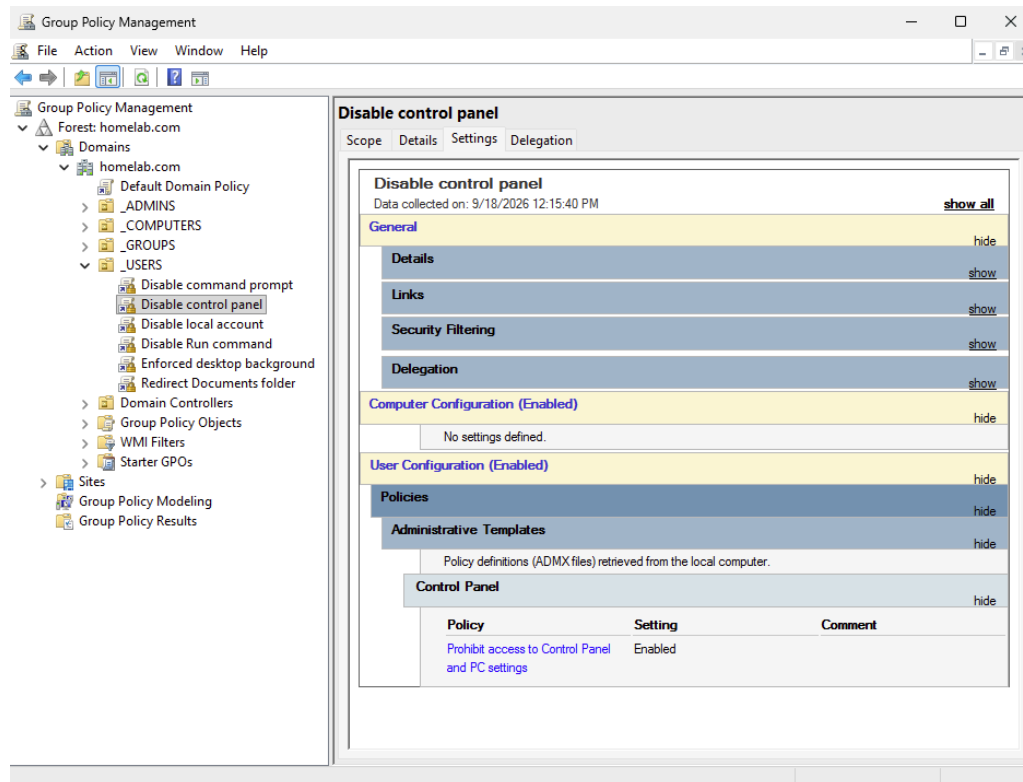


Objective 5: Create users and OUs for the domain to gain hands-on knowledge of user creation and management. My first interaction with the AD Users and Computers tool. I created a new OU called _USERS. Within that OU, I created a small handful of users. Here I learned how to reset passwords, unlock accounts, and also create Groups to incorporate users under, in a _GROUPS OU.



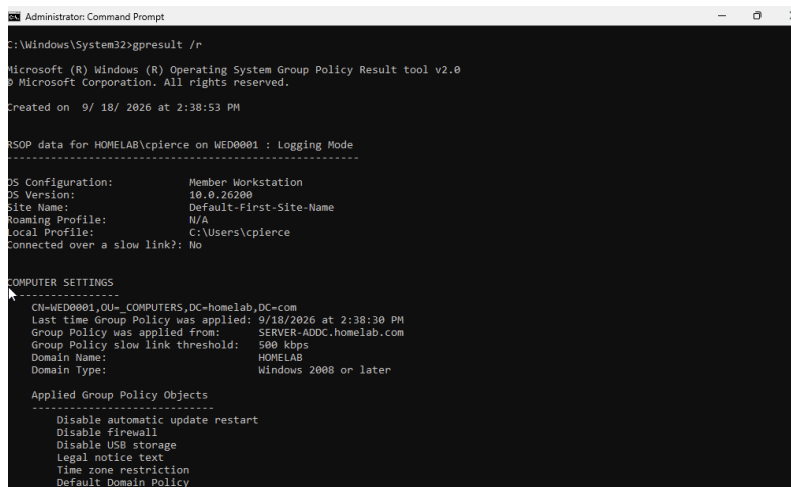
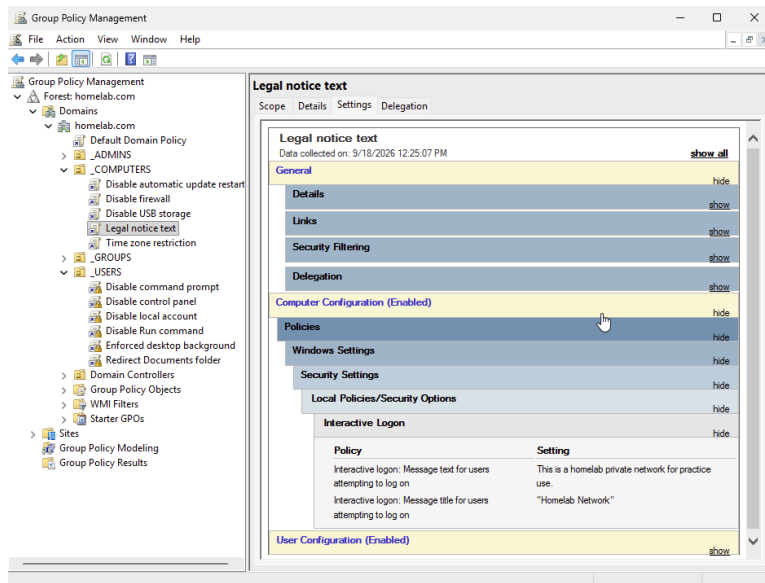
Challenges & Resolution: I did not know that it is good practice to create new OUs to move users into. After initially using the defaults, I created my own OUs.

Objective 6: Learn to apply GPOs to users. In the Group Policy Management utility, I learned how to create GPOs and link them to OUs and groups. It was as simple as a right click on the desired OU, and choosing to Create and Link a GPO. It was shocking how many configuration possibilities there are. There are so many that I would have to look up where to find them! I began creating User Configurations in the _USERS OU. I created policies to disable command prompt, disable Control Panel, disable local account, disable Run command, enforce a standard desktop background, and redirect the Documents folder to a shared folder.



Challenges & Resolution: The hardest policies to create were the enforced desktop background and redirecting the Documents folder. This is because these policies require setting up Shares, which I had never done before. I learned how to set up Shares, and while the Redirect Documents folder policy functioned as intended, I couldn't seem to get the Enforced desktop background policy to work. After going over my work over and over, I realized I was not specifying the correct path for the .jpeg I was enforcing as the desktop background. Even after I fixed that, I found my client VMs would still sometimes have a blank background. I found out two things: make sure the file path is correct, and make sure the server hosting the image is actually running!

Objective 7: Learn to apply GPOs to computers. Some policies I wanted enforced on the computer level rather than the user level. In the _COMPUTERS OU, I created policies for disabling automatic update restart, disabling firewall, disabling USB storage, displaying legal notice text before login, and a time zone restriction. The time zone restriction was the most complex as it required registry editing.



Challenges & Resolution: I could get my User policies to apply just fine, but none of my Computer policies would apply. I would run `gpupdate /force`, and then `gpresult /r` to see if the policies were active, but they just wouldn't activate. This is because, originally, I had the computers in the default Computers category. To the Group Policy Manager, they essentially didn't exist. I rectified this by creating a new _COMPUTERS OU, moving the computers into that OU, and creating and linking GPOs to that OU. All my policies were now being successfully enforced.

Conclusion: My Active Directory homelab was set up, making it an enriching learning experience. I could login as any domain user on any client VM, my VMs could communicate with each other as well as the outside Internet, and I had successfully created and enforced User-based GPOs and Computer-based GPOs.

Skills learned: password resets, account unlocking, directory alteration, user and computer creation, file sharing, Windows Server 2022/DHCP/DNS/AD DS setup, GPO creation and linking, registry editing, command line enforcement & evaluation of GPOs, VM creation, basic networking fundamentals of VMs.

